Nama: Hans Malvin Djojo

https://github.com/hansmalvin/LastMile

Insert Data:

```
test> use store
switched to db store
store> db.createCollection("products")
{ ok: 1 }
store> db.products.insertMany([
| { "_id": 1, "name": "Laptop", "category": "Electronics", "price": 1200, "quantity": 5, "tags": ["new", "powerful"] },
| { "_id": 2, "name": "Mouse", "category": "Electronics", "price": 25, "quantity": 50, "tags": ["wired", "ergonomic"] },
| { "_id": 3, "name": "Keyboard", "category": "Electronics", "price": 75, "quantity": 30, "tags": ["mechanical", "rgb"] },
| { "_id": 4, "name": "T-shirt", "category": "Clothing", "price": 20, "quantity": 100, "tags": ["cotton", "casual"] },
| { "_id": 5, "name": "Jeans", "category": "Clothing", "price": 60, "quantity": 75, "tags": ["denim", "casual"] },
| { "_id": 6, "name": "Book", "category": "Books", "price": 15, "quantity": 200, "tags": ["fiction", "novel"] },
| { "_id": 7, "name": "Headphones", "category": "Electronics", "price": 150, "quantity": 20, "tags": ["wireless", "noise-cancelling"] },
| { "_id": 8, "name": "Backpack", "category": "Accessories", "price": 40, "quantity": 60, "tags": ["durable", "waterproof"] },
| { "_id": 9, "name": "Smartphone", "category": "Electronics", "price": 900, "quantity": 15, "tags": ["5G", "android"] },
| { "_id": 10, "name": "Sweater", "category": "Clothing", "price": 45, "quantity": 80, "tags": ["wool", "warm"] }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5,
    '5': 6,
    '6': 7,
    '7': 8,
    '8': 9,
    '9': 10
  }
}
store> db.products.find()
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
```

```
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 3,
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    quantity: 30,
    tags: [ 'mechanical', 'rgb' ]
  },
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
```

```
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancell
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
    tags: [ 'wool', 'warm' ]
  }
]
```

Step 1

Equal price to 25

```
store> db.products.find({ price: { $eq: 25 } })
[
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  }
]
```

==

More than 100 pricenya

```
store> db.products.find({ price: { $gt: 100 } })
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  }
]
```

More and equal than 75 price

```
store> db.products.find({ price: { $lt: 50 }
[
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
```

price less than or equal to $20

```
store> db.products.find({ price: { $lte: 20 } })
[
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  }
]
```

price not equal to $40

```
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 3,
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    quantity: 30,
    tags: [ 'mechanical', 'rgb' ]
  },
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
```

```
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
    tags: [ 'wool', 'warm' ]
```

step 2

products where the category is either "Electronics" or "Clothing"

```
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 3,
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    quantity: 30,
    tags: [ 'mechanical', 'rgb' ]
  },
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
```

```
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling'
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
    tags: [ 'wool', 'warm' ]
  }
```

products where the category is not "Electronics" or "Clothing"

```
store> db.products.find({ category: { $nin: ["Electronics", "Clothing"] } })
[
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  }
]
```

step 3

products that are in the "Electronics" category AND have a price greater than $100

```
store> db.products.find({ $and: [{ category: "Electronics" }, { price: { $gt: 100 } }] })
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  }
]
```

products that are either in the "Books" category OR have a quantity less than 30

```
store> db.products.find({ $or: [{ category: "Books" }, { quantity: { $lt: 30 } }] })
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  }
]
```

products that are NOT in the "Clothing" category

```
store> db.products.find({ category: { $not: { $eq: "Clothing" } } })
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 3,
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    quantity: 30,
    tags: [ 'mechanical', 'rgb' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  }
]
```

products that are NEITHER in the "Books" category NOR have a price greater than $100

```
store> db.products.find({ $nor: [{ category: "Books" }, { price: { $gt: 100 } }] })
[
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 3,
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    quantity: 30,
    tags: [ 'mechanical', 'rgb' ]
  },
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
    tags: [ 'wool', 'warm' ]
  }
]
```

step 4

products that have both "new" AND "powerful" tags

```
store> db.products.find({ tags: { $all: ["new", "powerful"] } })
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  }
]
```

products where at least one element in the 'tags' array matches the specified criteria (starts with 'w')

```
store> db.products.find({ tags: { $elemMatch: { $regex: "^w" } } })
[
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
    tags: [ 'wool', 'warm' ]
  }
]
```

products that have exactly 2 tags

```
store> db.products.find({ tags: { $size: 2 } })
[
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 3,
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    quantity: 30,
    tags: [ 'mechanical', 'rgb' ]
  },
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
```

step 5

products that are in the "Electronics" category, have a price greater than $50, AND have a "wireless" tag

```
store> db.products.find({
| $and: [
| { category: "Electronics" },
| { price: { $gt: 50 } },
| { tags: "wireless" }
| ]
| })
[
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  }
]
```

products that are either in the "Clothing" category AND have a price less than $50, OR have exactly 2 tags.

```
  {
    _id: 1,
    name: 'Laptop',
    category: 'Electronics',
    price: 1200,
    quantity: 5,
    tags: [ 'new', 'powerful' ]
  },
  {
    _id: 2,
    name: 'Mouse',
    category: 'Electronics',
    price: 25,
    quantity: 50,
    tags: [ 'wired', 'ergonomic' ]
  },
  {
    _id: 3,
    name: 'Keyboard',
    category: 'Electronics',
    price: 75,
    quantity: 30,
    tags: [ 'mechanical', 'rgb' ]
  },
  {
    _id: 4,
    name: 'T-shirt',
    category: 'Clothing',
    price: 20,
    quantity: 100,
    tags: [ 'cotton', 'casual' ]
  },
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
  },
  {
    _id: 6,
    name: 'Book',
    category: 'Books',
    price: 15,
    quantity: 200,
    tags: [ 'fiction', 'novel' ]
  },
  {
    _id: 7,
    name: 'Headphones',
    category: 'Electronics',
    price: 150,
    quantity: 20,
    tags: [ 'wireless', 'noise-cancelling' ]
  },
  {
    _id: 8,
    name: 'Backpack',
    category: 'Accessories',
    price: 40,
    quantity: 60,
    tags: [ 'durable', 'waterproof' ]
  },
  {
    _id: 9,
    name: 'Smartphone',
    category: 'Electronics',
    price: 900,
    quantity: 15,
    tags: [ '5G', 'android' ]
  },
  {
    _id: 10,
    name: 'Sweater',
    category: 'Clothing',
    price: 45,
    quantity: 80,
```

Validation

```
store> db.products.find({
| $and: [
| { price: { $gte: 30, $lte: 80 }
| { tags: "casual" }
| ]
| })
[
  {
    _id: 5,
    name: 'Jeans',
    category: 'Clothing',
    price: 60,
    quantity: 75,
    tags: [ 'denim', 'casual' ]
  }
]
store>
```